SCHOOL OF
BUSINESS MANAGEMENT

A REPORT ON

Dynamic Phase-Wise Analysis and Visualization of IPL Matches

For:

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Submitted By

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2. Preface

In today's age of data-driven decisions, the use of analytics in sports has revolutionized how teams train, play, and plan. Among sports, cricket, especially in forms like the Indian Premier League (IPL), has seen a remarkable transformation, with data analytics becoming pivotal in player analysis and tactical decision-making. Having a passion for cricket and an increasing interest in data science, I was attracted to the prospect of exploring how real-time data would have an impact on decisions made on the field. Such a convergence of interest and prospect motivated me to take a summer internship in Cricket Analytics at Ball Track Analyzer.

The topic of this internship was selected after a thorough examination of its significance in today's sporting environment. Cricket, as a sport of close margins and constantly changing trends, yields enormous amounts of information—from each ball delivered to each stroke hit. As the intricacy and intensity of such tournaments as the IPL increase, teams are dependent on detailed statistics and visual narratives in order to get a tactical edge. This brought to the fore the requirement for professionals who could take raw data and turn it into actionable information. Cricket analytics, apart from necessitating statistical expertise and technical proficiency, needs an understanding of the sport itself—making it a challenging and mentally stimulating task. During the course of my internship, I had the opportunity to work on real-time IPL data, analysing over **120,000 ball-by-ball records** using *BallTrack Analyzer*, a powerful tool designed to handle large datasets with high efficiency. My work involved computing more than **10 key performance indicators (KPIs)**, such as economy rate, strike rate, false shot percentage, and more, segmented across the three primary match phases: **Powerplay, Middle Overs, and Death Overs**. These KPIs were calculated for both teams in each match to draw comparisons and highlight match-defining moments.

In addition to numerical analysis, I also used visual representation of data, which is essential in communicating insights to non-technical stakeholders. I created 25+ visualizations such as pitch maps and wagon wheels to represent player performance zones and shot distributions. To provide a dynamic take on the analysis, I produced animated match flow graphs capturing momentum changes using storytelling tools like Flourish Studio.
This internship permitted me to connect theory to practice—using Python to manipulate data, statistical thinking to draw conclusions, and data visualization methods to portray conclusions in a meaningful way. What is more important, however, is that it taught me to appreciate the increasing relevance of sports analytics to real-world decision-making.
From this experience, I not only gained a deeper insight into cricket from a data perspective but also developed my analytical skills, attention to detail, and problem-solving skills. This project again confirmed my interest in choosing a career as a data scientist, especially for fields that marry passion and analytical depth, such as sports technology and performance analysis.

Durgapur, West Bengal *Sayan Mukherjee*
29/05/2025

3. Acknowledgement

I would like to use this moment to write with all my heart a thank you note to everyone who guided and supported me during the period of my internship at Ball Track Analyzer.

I am first and foremost grateful to Mr. Saiprasad Kagne, Founder and CEO of BallTrack Analyzer, for providing me with the precious chance to serve as a Cricket Analytics Intern under his guidance. His vision, direction, and continuous encouragement were a critical part of developing my knowledge on sports analytics and enabled me to delve into live cricket data with richness and intent. I am particularly thankful for his insights, invaluable feedback, and the trust he had on me during the internship.

I also want to thank the whole BallTrack Analyzer team for providing a productive and stimulating environment to work in that enriched my learning process immensely. Their encouragement, suggestions, and knowledge sharing helped me enjoy this experience as much as I learned from it.

Lastly, I want to thank my fellow students, teachers, and family members for their constant support, which kept me going with the project.

This internship has also been an important milestone in my studies and working life, and I am more than grateful to all those people who made it successful.

Sayan Mukherjee

4. Executive Summary

This BallTrack Analyzer internship was an in-depth exploration of the new field of sports analytics, specifically in the intense setting of Indian Premier League (IPL) cricket. With data now at the centre of performance analysis in contemporary sports, the internship sought to process real-time match data and extract actionable insights that can be used by teams in tactical planning, player evaluation, and match strategy.

During the internship, over 120,000 ball-by-ball IPL records were processed with BallTrack Analyzer, an in-house proprietary analytical engine specifically designed for analysing large-scale sports data within seconds. The principal analytical activity consisted of extracting and interpreting 10+ performance indicators (KPIs) like economy rate, strike rate, false shot rate, dot ball rate, and boundary frequency. These were computed phase-wise—Powerplay (1–6 overs), Middle Overs (7–15), and Death Overs (16–20)—for both batting and bowling teams in several IPL games.

These findings were also made interactive with 25+ custom visualizations, such as pitch maps, wagon wheels, heat zones for the players, and animated graphs of match momentum over time. These allowed for simple comprehension of the players' performance in various conditions of a match and of the changes in momentum during the game. Interactive narratives and animated videos facilitated a new level of analysis, particularly in marking turning points and pressure moments.

Key Findings:

- Middle overs bowling effectiveness, supported by steady dot-ball tactics, was frequently an indicator of control of the match.
- Death overs exhibited dramatic differences in false shot % and boundary leakage, emphasizing the necessity of unique end-overs tactics.
- Certain batsmen revealed high-scoring areas on wagon wheels, indicating field placement tendencies to be exploited.
- Momentum graphs provided obvious points of inflection where game directions were changed by tactical alterations (for example, bowling alternations or batting changes).

Recommendations:

•Phase-Specific Role Deployments: Use bowlers and batsmen in the match phases when their KPIs show peak performance.

•Optimization of Bowling Strategy: Tailor bowling lengths and angles with pitch map analysis for specific player match-ups.

• Tactical Application of Visualizations: Embed animated images into coaching debriefs and strategy meetings to enhance the reporting of data-based insights.

• Predictive Match Flow Tracking: Leverage momentum changes to foresee decision-making situations (e.g., when to call timeouts or substitute essential players).

The project not only added to match-level analytical findings but also became a stepping stone in seeing where data science, visualization, and cricket strategy could meet. It reinforced the increasing role of analytics in sports and reestablished the importance of marrying domain knowledge with technical ability. The internship was a major leap in translating academic learning into a real-world, high-pressure setting.

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7. Introduction

Cricket has also seen a dramatic shift in recent years with the popularity of T20 formats changing the way the game is played, watched, and analyzed. And of these, the Indian Premier League (IPL) is the most commercially lucrative and strategically sophisticated competition, with a combination of international stars, data-driven strategy, and quick game development. In an environment as pressure-filled and as quick-moving as the current one, old-fashioned scorecards and simple statistics cannot do justice to the complexity of match flow, player performance, and tactical changes. Advanced, phase-specific analysis of performance is now more important than ever—not merely for analysts and coaches, but for broadcasters, spectators, and even players themselves.

This project, "Dynamic Phase-Wise Analysis and Visualization of IPL Matches," is a endeavor to satisfy this requirement by splitting the game into its three phases—Powerplay (overs 1–6), Middle Overs (7–15), and Death Overs (16–20)—and performing a in-depth performance analysis of each team over each of these intervals. The project seeks to break away from statistical generalities by examining the dynamic evolution of the game across phases, and how the players learn to adapt—or not—according to changing conditions, roles, and oppositional strategies. The phase-wise analysis gives us a tactical perspective on the game, real match strategies employed by teams in action.

To process and handle the enormous amount of data available in the IPL, which had more than 120,000 individual deliveries, a high-performance utility called BallTrack Analyzer was employed in the project. The utility facilitated real-time parsing, filtering, and computation of several critical KPIs like economy rate, strike rate, dot ball rate, false shot rate, boundary frequency, and the like. These statistics were then compared phase by phase for major players—both batsmen and bowlers—across several matches to reveal patterns, tendencies, and vulnerabilities that would otherwise remain undetectable through traditional scorecards.

What sets this project apart is its disproportionate use of visualization as a medium of narrative and strategic information. More than 25 distinct visual outputs were generated, including wagon wheels, pitch maps, heat zones, and animated match momentum graphs. These representations not only provide insights into the analytical results but also assist in identifying spatial patterns (such as scoring zones or lengths to bowl) and pressure moments, as well as changes in momentum. Such as, wagon wheels assisted in determining reliable scoring zones for aggressive batsmen in the Death Overs, while pitch maps indicated effective bowling lengths that suppressed runs in the Powerplay. Graphic representation of animated momentum revealed when and how match pathways altered—offering key insights into captaincy choices, bowling switches, or batting meltdown.

Apart from equipping with analytical capabilities, the internship experience also solidified the application of data science practices in real-time sporting usage. Python was predominantly applied for data wrangling, KPI calculation, and match phase filtering. The project further entailed structured report writing, comparative analysis, and insight synthesis—abilities imperative in the field of sports intelligence and performance analytics.

On a larger scale, this internship demonstrates the utility of phase-by-phase analysis as a planning instrument. Teams can utilize such knowledge to refine their rotations of bowlers, control risk exposure over overs, and condition their batting orders to target weaknesses in the opposition. Furthermore, such granular analyses can be used to simulate matches, scout

players, and make auction decisions, providing long-term benefits that transcend the limits of a single year.

Finally, the project remains faithful to the title—Dynamic Phase-Wise Analysis and Visualization of IPL Matches—by not only accepting the dynamic and changing essence of T20 cricket but also presenting a methodical way to dissect it into intelligible and actionable phases. It highlights how information, when utilized intelligently and contextually, can reveal the subtleties of the game and open the door for smarter, quicker, and wiser cricketing decisions.

8. Methodology

Research Objective

The aim of this study is to conduct a dynamic, phase-wise performance analysis of IPL matches from real-time ball-by-ball data. Utilizing Python for data processing, BallTrack Analyzer for high-speed data extraction, and Flourish Studio for interactive story-telling, the study will calculate key performance indicators (KPIs)—such as economy rate, strike rate, dot ball percentage, and false shot percentage—during the Powerplay, Middle, and Death overs. The aim is to produce actionable insight into player and team strategy through both statistical inference and compelling visualization.

Research Methodology

Primary Data

Universe:
The first universe of the study consists of all IPL matches examined during the internship, with a specific focus on more than 120,000 ball-by-ball records over several seasons of IPL.

Instrument:
The most important tools and instruments for data extraction and analysis are:

BallTrack Analyzer for large-scale, real-time data parsing

Python (Pandas, NumPy, Matplotlib) for computation of KPI and statistical analysis

Flourish Studio for building interactive and animated visuals

Sample Size:
The data set contains over 120,000 deliveries from several matches, players, and match segments (Powerplay, Middle Overs, Death Overs). Every delivery has features such as batsman, bowler, runs scored, delivery type, dismissal, and shot result.

Secondary Data
Sources:
Secondary data helped to cross-validate insights and reinforce the primary dataset. These sources were:

Official IPL scorecards and commentary logs

Cricbuzz and ESPNcricinfo databases for cross-verifying player stats

Research papers and online articles on cricket analytics for conceptual grounding

GitHub repositories and forums for optimization of code and visualization approaches

This approach guaranteed a sturdy and robust methodology for analyzing IPL match statistics, calculating phase-wise KPIs, and deriving action-driven insights through the marriage of technical analysis and intuitive visual narratives.

Assumptions

1- Accuracy of Source Data:
We assume that the ball-by-ball IPL data utilized from BallTrack Analyzer and official scorecards is accurate, exhaustive, and free from error in recording.

2- Phase Division Consistency:
The phases of the match are divided evenly as follows: Powerplay (Overs 1–6), Middle Overs (7–15), and Death Overs (16–20), regardless of the situation of the match or interruptions.

3- Uniform Conditions:
Analysis uses the hypothesis that pitch and weather conditions for a match period are close to constant, such that variations in performance are assigned to player strategy and execution as the main determinants.

4- Independent Deliveries:
Every delivery is treated as an independent data point for KPI calculation, though real match situation (e.g., pressure, situation of the game) can potentially affect player actions.

5- No External Bias:
All matches are presumed to have been played honest and free of external influences like umpire mistakes or disputed calls.

6- Static Role Definitions:
The players were compared in terms of their role definition (batsman or bowler) without considering hybrid roles or dynamic game tasks.

Limitations-

1- Data Coverage**:**
The dataset, while extensive, includes only selected IPL matches and may not represent the full spectrum of playing conditions or strategies across all seasons.

2- Contextual Factors Ignored**:**
External factors like pitch behaviour, weather conditions, crowd influence, or match pressure were not explicitly factored into the analysis due to data unavailability.

3- Static Phase Boundaries
Match phases were divided using fixed over ranges, which may not align with dynamic shifts in momentum or intent during actual gameplay.

4- Player Roles Are Generalized**:**
Players were analysed based on broad roles (e.g., batsman, bowler) without accounting for specific game situations like pinch-hitting or part-time bowling spells.

5- Tool Constraints**:**
Visualizations and analysis were limited by the functionalities of the tools used (e.g., BallTrack Analyzer, Flourish Studio), which may have restricted deeper contextual modelling.

6- Lack of Predictive Modelling**:**
The project focuses on descriptive analytics rather than predictive insights, such as forecasting player performance or match outcomes.

9. Findings

->Performance Breakdown | Jos Buttler's 97 vs DC: A Masterclass in T20 Batting

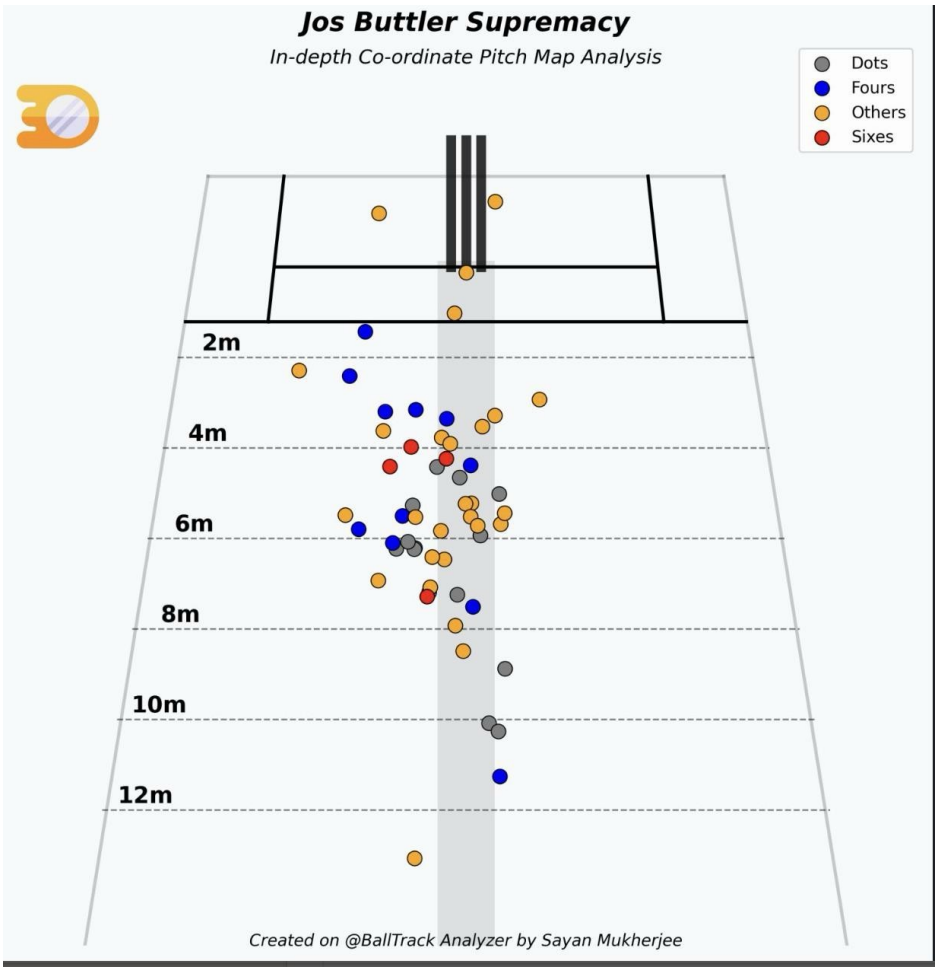


Figure-9.1 - Jos Buttler's Pitch Map

The Jos Buttler's innings of 97 (54) against Delhi Capitals pitch map visualization gives key insights into bowlers' targeted areas and the way Buttler counteracted these lengths.
Pitch Map Findings:

Ball Distribution:

Most of the balls were pitched in the good length area (4m–7m), a common strategy to restrict hard-shots.
Buttler counteracted this strategy with intelligent stroke play, turning even good-length balls into boundary shots.
Shot Outcomes:

Dot balls (Blue): Aggregations of dot balls were primarily on back-of-a-length and outside off-stump, indicating regions where Buttler was briefly contained.

Boundaries (Red & Yellow): There was an extremely high concentration of fours and sixes in the 6–7m area, particularly when bowlers went out of line (middle and leg).
Sixes (Gray): Crucially, Buttler passed the boundary on fuller deliveries, particularly in the 4m–6m region, highlighting his effectiveness at punishing over-pitched balls.
Bowling Impact Zones:

Bowlers seldom probed the short-pitched area (8m–10m), perhaps in terror of Buttler's pull and hook capability.
Key Insights:
Jos Buttler showed better control over good length balls and punished every flaw in line or length. The analysis suggests that a more appropriate combination of short balls and wide yorkers could have limited his run flow of boundaries.

→How KKR Decoded & Dominated SRH's Bowling Attack: A Deep-Dive Analysis

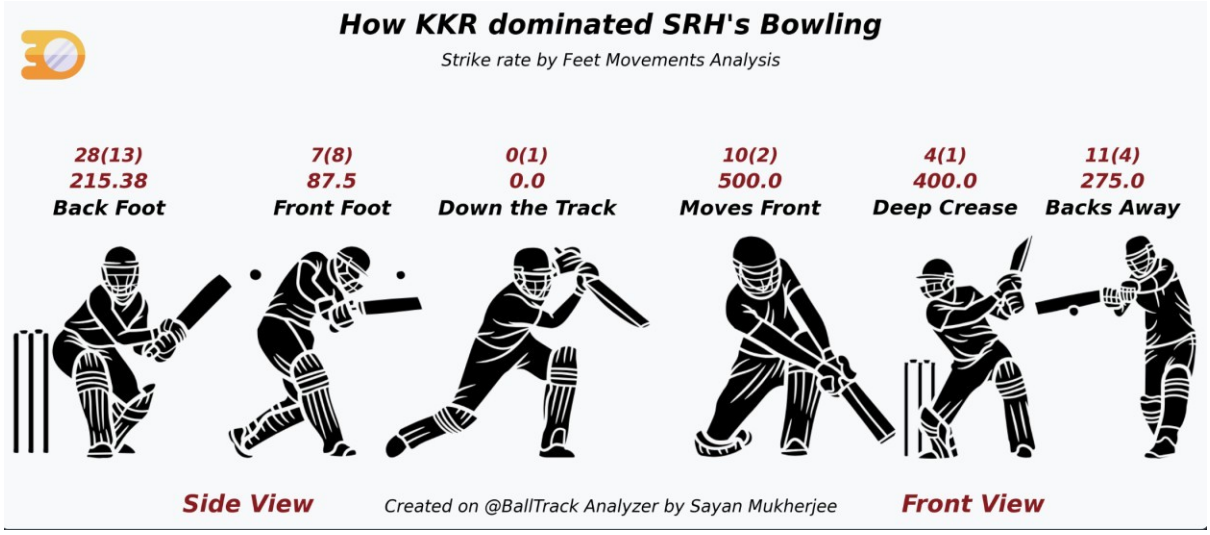


Figure-9.2 - Feet Movement Analysis

Feet Movement Analysis

28 runs were scored off 13 balls by the back foot batters with a strike rate of 215.38, showcasing dominance against good-to-short-length balls.

Far too high strike rates were also achieved for:

Moves Front: 500.0 (10 off 2)

Deep Crease: 400.0 (4 off 1)

Backs Away: 275.0 (11 off 4)

Little output was made on the front foot (87.5 SR), and no runs were scored when batsmen stepped down the track, suggesting a strategic choice of remaining deep in the crease and playing late.

The fluctuating movement trends mirror KKR's skill at controlling depth and angles in the crease to negate SRH's pace attack.

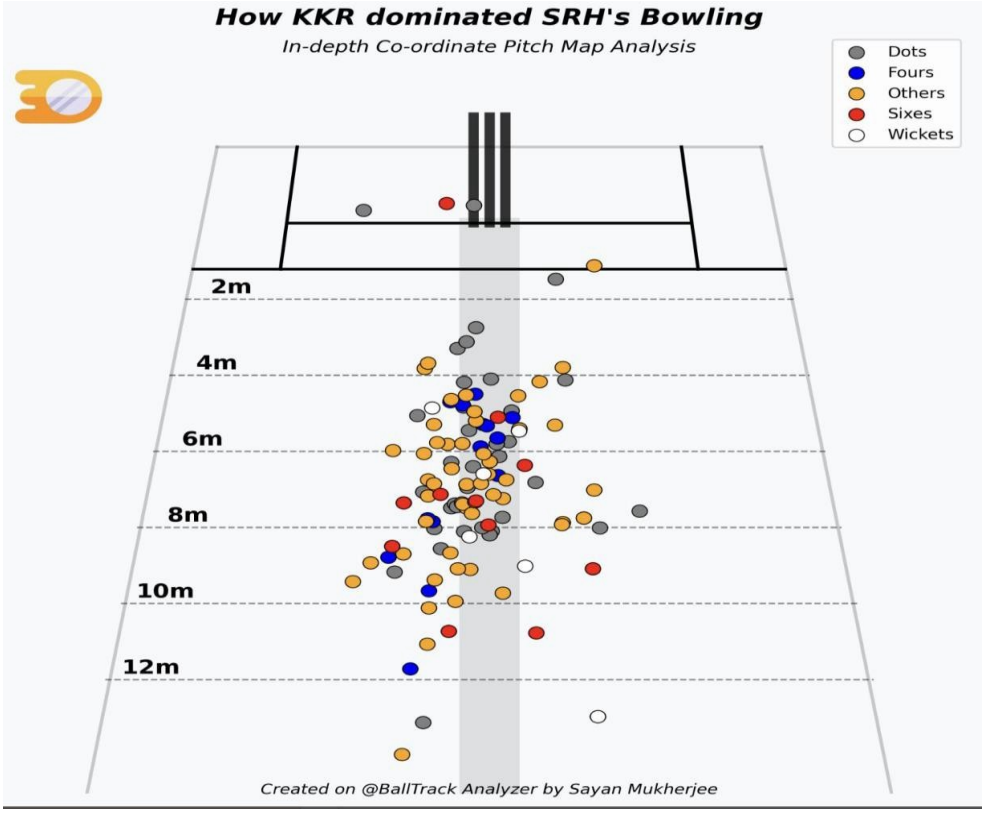


Figure-9.3 - Pitch Map of KKR

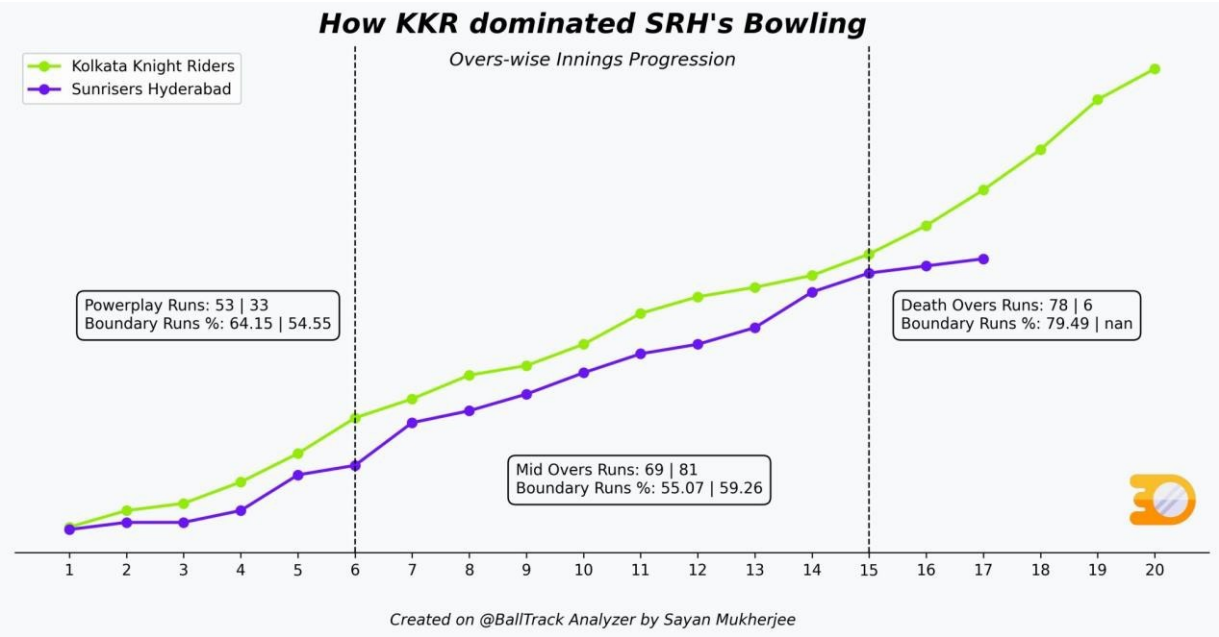
Pitch Map Analysis

KKR batsmen focused primarily on the 6m to 10m length areas, turning good-length balls in the past into boundary-scoring shots.

A high concentration of boundaries (red and yellow marked) was seen in this corridor, suggesting free-flowing back-foot and crease-based hitting.

SRH bowlers were not able to change the lengths appropriately, persistently giving feeds to KKR's hitting area and permitting the batsmen to overpower them with minimal line or length change.

The gray markers for wickets were predominately in the range of 4m–6m, indicating fuller lengths were relatively more dangerous but not being exploited



Graph-9.4 - Overwise Progression Chart

Overwise Progression Analysis:

Powerplay Phase (Overs 1–6):

KKR posted 53 runs with a boundary run percentage of 64.15%, which pointed to initial momentum.

SRH could score only 33 runs at 54.55%, trailing behind with early aggression.

Middle Overs (7–15):

SRH led briefly (81 runs against KKR's 69), with a marginally higher boundary percentage (59.26% against 55.07%).

SRH couldn't translate this mid-phase lead into long-term momentum, though.

Death Overs (16–20):

KKR showed explosive finishing, with 78 runs scored, and a boundary contribution of 79.49%—a decisive component of their finishing ability.

SRH scored only 6 runs, reflecting a disastrous collapse and inability to finish off well.

Summary of Key Findings:

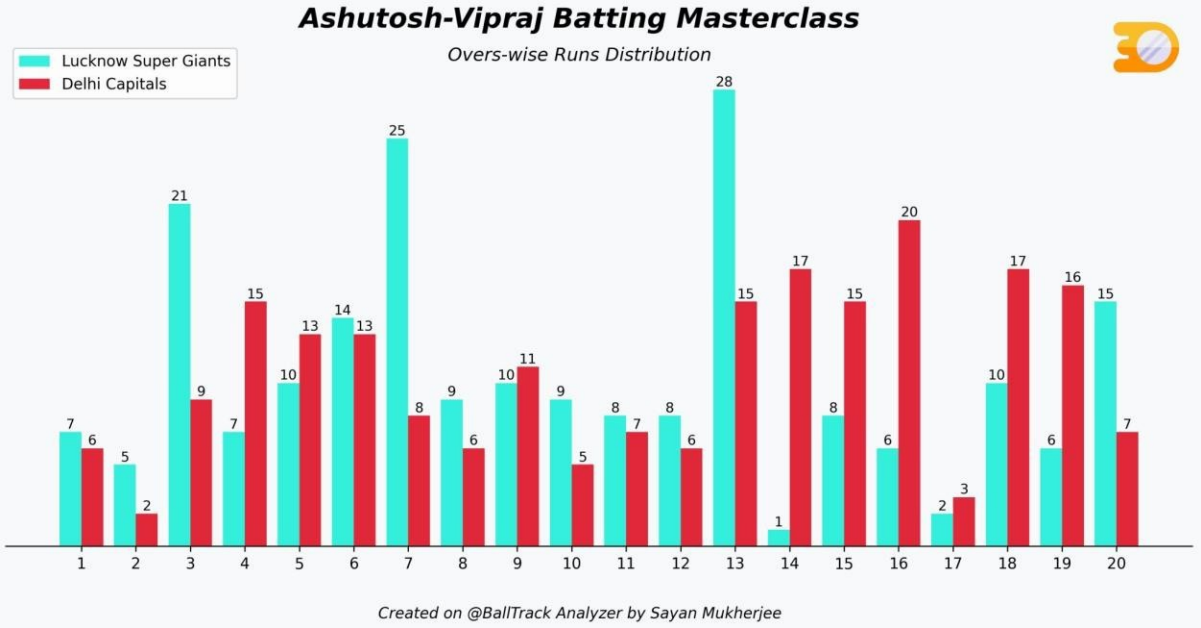
1- Length Exploitation: KKR batsmen consistently attacked the 6–10m length area, turning regular deliveries into high-scoring shots.

2- Crease Intelligence: Good strike rates off crease-based movements indicated a solid tactical base in reacting to length and pace.

3- Phased Dominance: KKR gained dominance in powerplay and death overs, gaining momentum bookends and scoreboard pressure.

4- SRH's Tactical Failures: Failure to evolve bowling lengths and not having finishing strength weighed heavily against them.

->How Delhi Capitals Cracked Lucknow's Code: Tactical Precision Meets Data-Driven Execution



Graph-9.5 - Over-Wise Scoring Breakdown

Over-wise Scoring Breakdown

LSG's innings contained two broad spikes:

Over 3: 21 runs

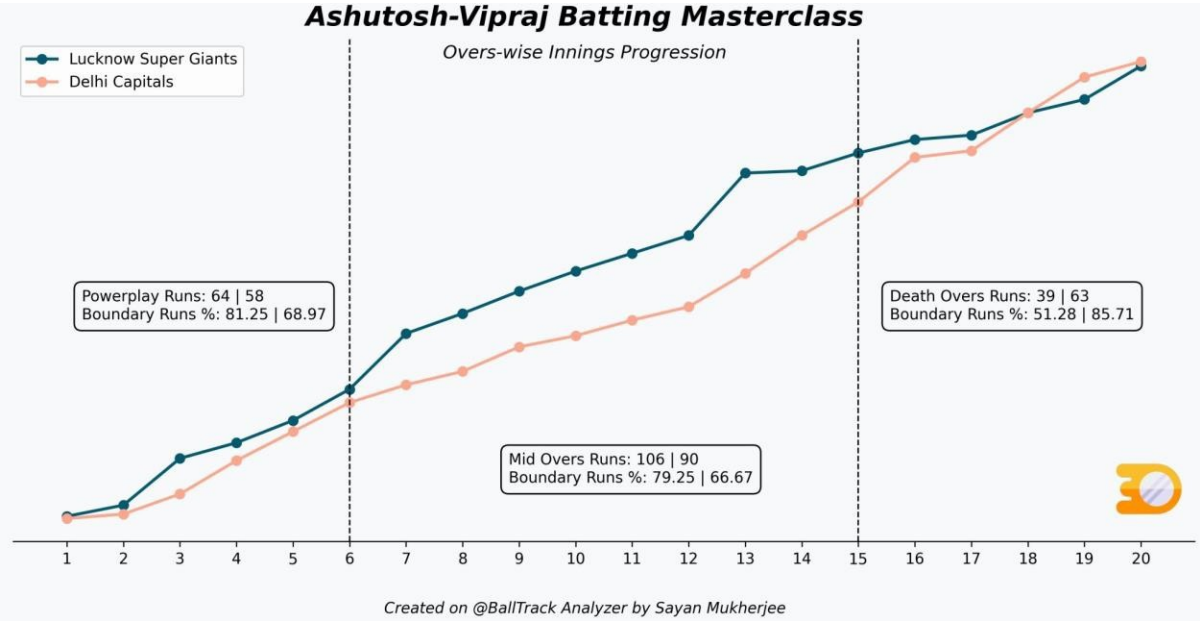
Over 13: 28 runs
These were primary momentum builders in an otherwise constant scoring curve.

DC's momentum burst arrived after the 14th over, prominently:

17th: 20 runs

18th & 19th: 17 and 16 runs respectively

Insight: DC's innings was more back-loaded, whereas LSG emphasized aggressive beginnings — revealing a different tempo approach between the two teams.



Graph-9.6 - Phase of Momentum Graph

Overs-wise Innings Progression: Phases of Momentum
Powerplay Surge:

LSG: 64 runs (81.25% through boundaries)

DC: 58 runs (68.97% through boundaries)
LSG took the lead early with boundary-laden aggression.

Middle Overs Domination:

LSG scored 106 runs with 79.25% boundary contribution — a phase where Ashutosh and Vipraj harvested matchups.

DC trailed at 90 runs, with just 66.67% through boundaries.

Late-Innings Shift:

DC overcame LSG 63 for 39 in the last 5 overs with a whopping 85.71% boundary share, turning late momentum in their favor.

LSG's finish slowed down with greater dependence on running between wickets (boundary % fell to 51.28%).

Insight: LSG established an overpowering platform through the first 15 overs — but their death-over implementation went awry, enabling DC to bridge the gap considerably.

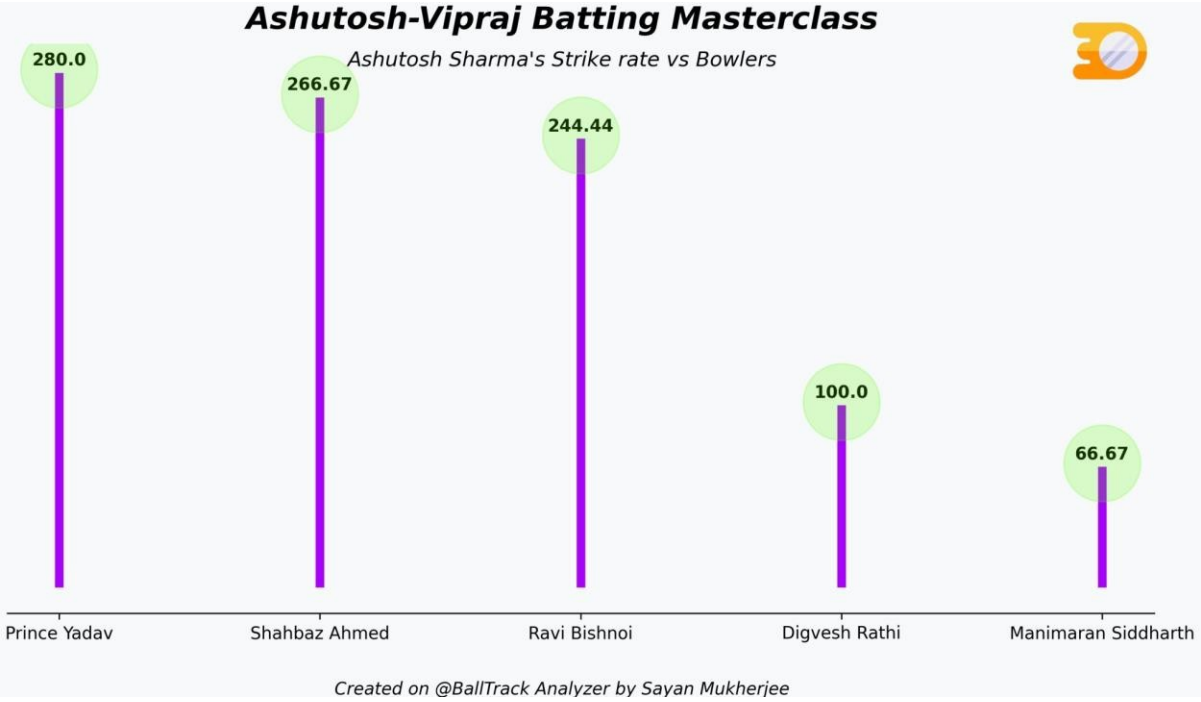


Figure-9.7 - Match -up Acceleration Graph

Ashutosh Sharma's Matchup-Based Acceleration

Ashutosh's runs were extremely matchup-specific, delivering maximum effect against bowlers with poorer control or pace:

280.00 strike rate against Prince Yadav

266.67 against Shahbaz Ahmed

244.44 against Ravi Bishnoi

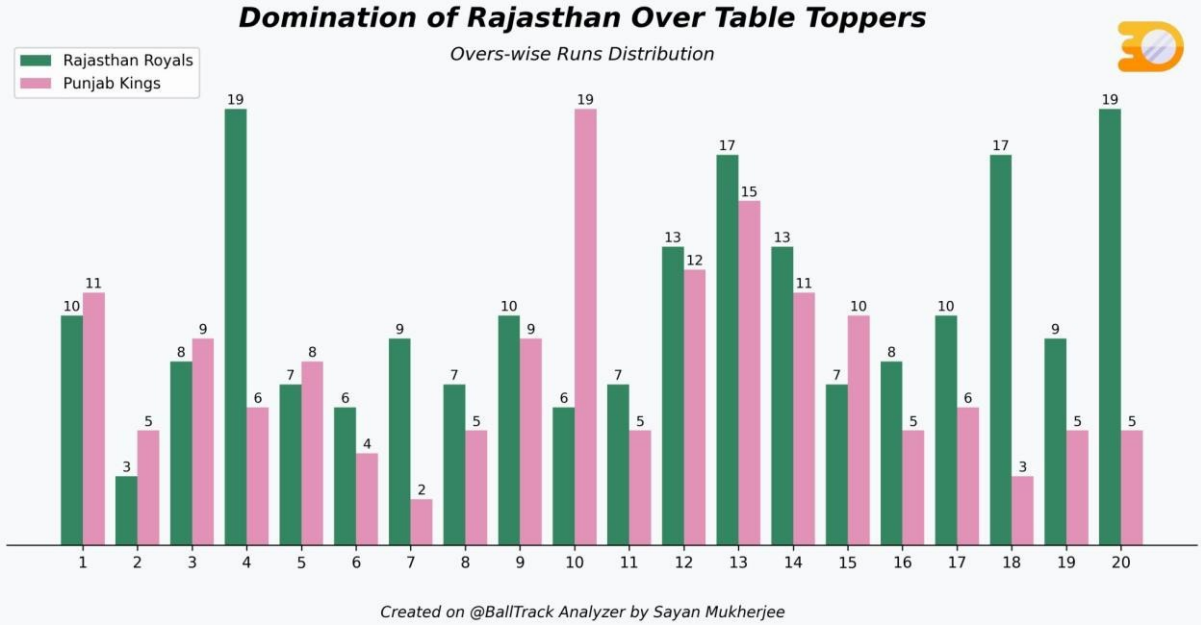
A sharp drop in scoring was observed against:

Digvesh Rathi (100.0)

Manimaran Siddharth (66.67)

Observation: Ashutosh adopted a definite risk-reward approach — aggressively playing medium pace and spin, while conserving his wicket against tight bowlers.

->Domination of Rajasthan Royals over Table Topper’s(Punjab Kings)



Graph-9.8 - Over-Wise Bar Chart

The overs-wise bar chart shows Rajasthan Royals outscoring Punjab Kings in the majority of overs throughout the innings:

RR scored 19 runs in each of the 4th, 13th, and 20th overs, which means aggressive cricket in transition and death overs.

PBKS scored their maximum of 19 runs in the 10th over but could not maintain the momentum after that.

RR surpassed PBKS in 14 out of 20 overs, showing consistency in putting pressure.

Observation: Rajasthan exhibited better rotation of the strike and boundary-hitting rate, particularly in key overs.

Shot Effectiveness by Length and Line

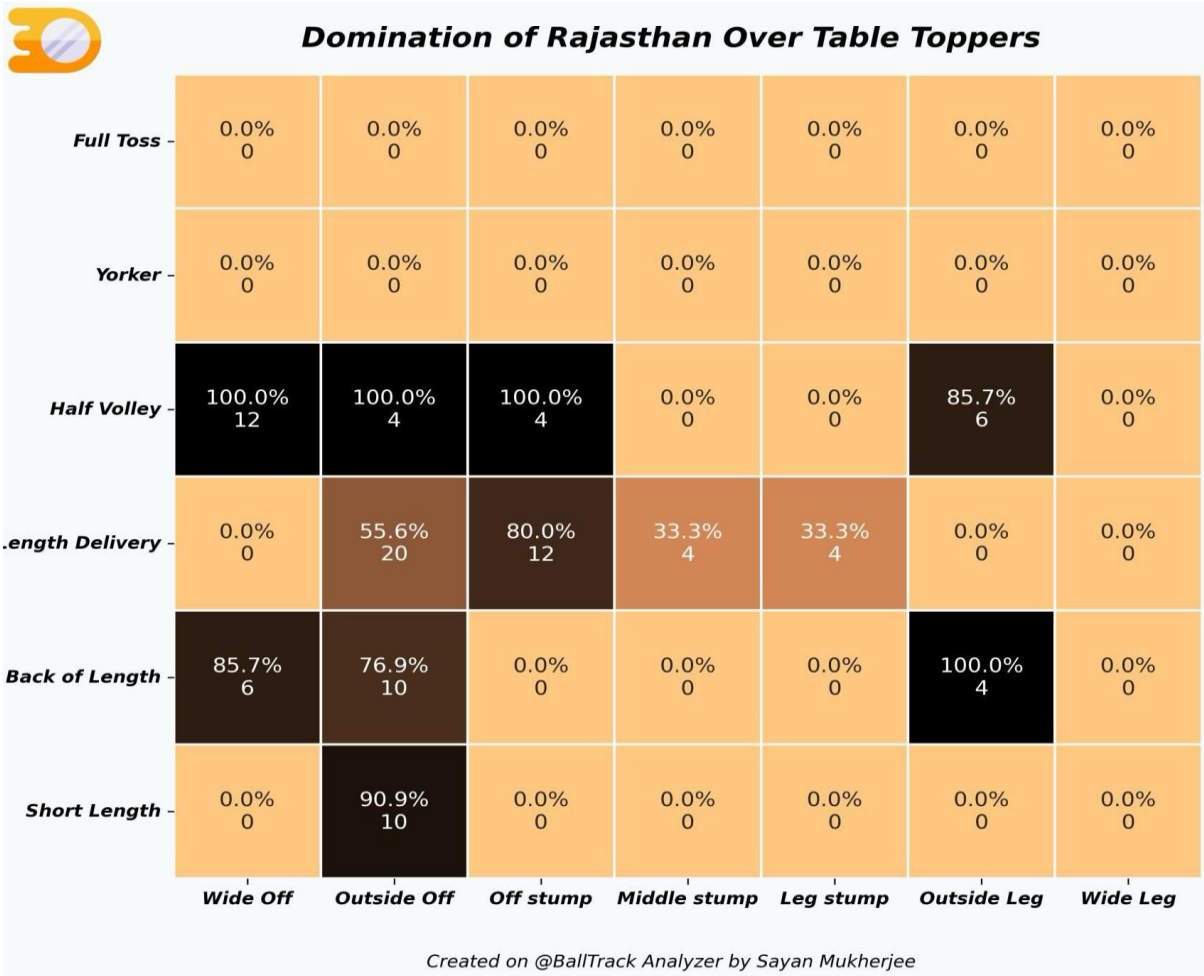


Figure-9.9.1 - Pitch Zone Heatmap

The pitch zone heatmap analyzes the batting success rate of Rajasthan batsmen against various types of deliveries and lines of attack:

There was 100% scoring success for offside zone (wide/off stump) half-volleys.

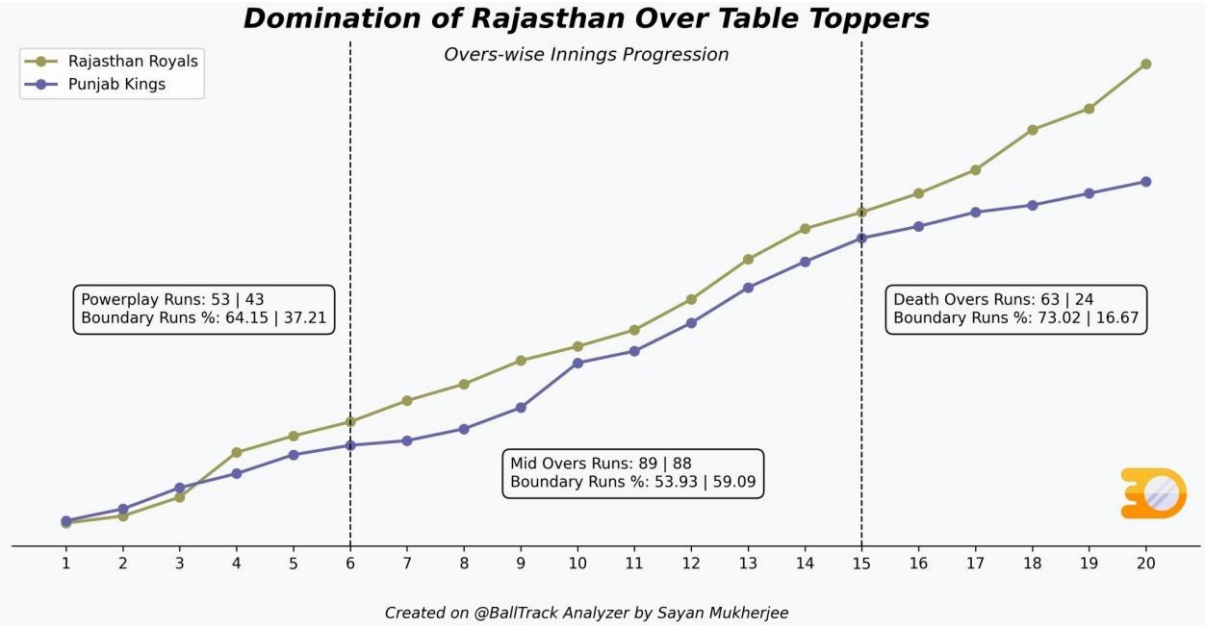
Back of length and short balls outside off stump had good yields — 85.7%, 76.9%, and 90.9% scoring rates, respectively.

Yorkers and full tosses were non-existent, reflecting a tactical failure in PBKS's death bowling arsenal.

RR did not score off leg-stump or wide-leg deliveries, with little exploitation of that line by PBKS bowlers.

Explanation: Rajasthan batsmen were geared up to attack specific lengths (slot and short outside off), while Punjab's attack was unable to challenge them with stumps-targeting or fuller-length options.

Over Wise Cumulative Progression



Graph 9.9.2 – Phase Wise Analysis of the Game

Phase	RR Runs	PBKS Runs	RR Boundary %	PBKS Boundary %
Powerplay (1–6)	53	43	64.15%	37.21%
Middle Overs (7–14)	89	88	53.93%	59.09%
Death Overs (15–20)	63	24	73.02%	16.67%

Key Insights:

Rajasthan gained early control in the powerplay and sustained the run rate in the middle overs.

The death overs were the turning point, wherein RR scored 63 runs—a 39-run difference against PBKS.

The batting approach of RR was predominantly boundary-oriented in both the powerplay and death overs.

10. Recommendation

This research, based on live IPL data and cutting-edge visual analytics, has provided various strategic insights into match flow, player performance, and tactical decision-making. In light of the findings and observed opportunities, the following are recommended to teams, analysts, and researchers in the future:

1. Tactical Player Deployment by Match Phase

Teams should deploy players according to their phase-wise strengths as per their KPIs:

Use bowlers with poor economy and dot ball percentage in Middle Overs to keep control.

Use explosive batsmen in Death Overs where boundary rate and tolerance of false shots is greater.

Phase-specific match-ups can optimize effect and minimize risk.

2. Data-Driven Bowling Strategy Adjustments

Use pitch maps and heat zone analysis to:

Detect lengths in which opposition batsmen have lower scoring success.

Adjust bowling lines and angles according to batsmen wagon wheel statistics (e.g., not giving slot balls to batsmen such as Jos Buttler).

Vary lengths, particularly during Death Overs, in order not to present predictable attacking zones as seen during match collapses.

3. Infuse Visual Analytics into Coaching Processes

Utilize animated graphs, pitch maps, and shot zone graphics during:
Pre-match planning to evaluate opposition patterns.

Post-match analysis to review wasted tactical chances.

Strategy sessions to enhance analyst-coaching dialogue.

4. Refine Finishing Strategies with Momentum Flow

The momentum graphs during matches clearly show how final overs tend to set the game in motion. Teams should:
Utilize real-time metrics to decide in-game like when to go for a power hit or play cautiously.

Time bowling power-hitting sessions or tactical replacements against momentum patterns.

5. Enhance Death Over Planning

Major collapses (e.g., SRH vs KKR) were caused by:

A lack of flexibility in bowling lengths.

Inadequate field placement knowledge.

Lack of batting firepower.

Both bowling options (yorkers, slow bouncers, wide lines) and field configurations must therefore be pre-simulated using analytic tools.

6. Future Research and Tool Improvement

Increase Data Context: Add pitch condition, match pressure, weather, and crowd information to increase model strength.
Add Predictive Models: Apply machine learning to predict player performance under various match conditions.

Tool Improvement: Improve tools such as BallTrack Analyzer to enable richer contextual labelling (e.g., player intent, game state) and live adaptability.

7. Phase-wise Auction and Scouting Suggestions

Teams can recruit players based not just on averages or strike rates but also on their:
Success in particular stages.

Advantages against certain match-ups based on past encounter patterns.

This can allow for more intelligent auction tactics and more balanced squads.

11. References

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12. Annexures

Annexure A: Sample Ball-by-Ball Data Snapshot

Date	Batting Team	Bowling Team	Striker	Non-Striker	Bowler	Innings	Unique	Over Number	Over Ball
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Priyansh Arya	Prabhsimran Singh	Mohammed Shami	1	0.1	1	1
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Prabhsimran Singh	Priyansh Arya	Mohammed Shami	1	0.2	1	2
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Prabhsimran Singh	Priyansh Arya	Mohammed Shami	1	0.3	1	3
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Prabhsimran Singh	Priyansh Arya	Mohammed Shami	1	0.4	1	4
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Prabhsimran Singh	Priyansh Arya	Mohammed Shami	1	0.5	1	5
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Prabhsimran Singh	Priyansh Arya	Mohammed Shami	1	0.6	1	6
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Priyansh Arya	Prabhsimran Singh	Pat Cummins	1	1.1	2	1
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Priyansh Arya	Prabhsimran Singh	Pat Cummins	1	1.2	2	2
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Priyansh Arya	Prabhsimran Singh	Pat Cummins	1	1.3	2	3
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Priyansh Arya	Prabhsimran Singh	Pat Cummins	1	1.4	2	4
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Priyansh Arya	Prabhsimran Singh	Pat Cummins	1	1.4	2	5
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Prabhsimran Singh	Priyansh Arya	Pat Cummins	1	1.5	2	6
13-04-2025 19:30	Punjab Kings	Sunrisers Hyderabad	Prabhsimran Singh	Priyansh Arya	Pat Cummins	1	1.6	2	7

Figure-10.1 -Screenshot of Dataset 1

Runs	Wides	Team Runs	Remaining Overs	Lead By	Trail By	Fall Of Wickets	Day	Session	Leg Byes	No Balls	Balls	Byes	Remaining Balls	Remaining Runs
1	0	1	19.5	0	0	0	1	1	0	0	1	0	119	
1	0	2	19.4	0	0	0	1	1	0	0	2	0	118	
0	0	2	19.3	0	0	0	1	1	0	0	3	0	117	
4	0	6	19.2	0	0	0	1	1	0	0	4	0	116	
4	0	10	19.1	0	0	0	1	1	0	0	5	0	115	
4	0	14	19	0	0	0	1	1	0	0	6	0	114	
0	0	14	18.5	0	0	0	1	1	0	0	7	0	113	
6	0	20	18.4	0	0	0	1	1	0	0	8	0	112	
4	0	24	18.3	0	0	0	1	1	0	0	9	0	111	
1	1	25	18.3	0	0	0	1	1	0	0	9	0	111	
1	0	26	18.2	0	0	0	1	1	0	0	10	0	110	
0	0	26	18.1	0	0	0	1	1	0	0	11	0	109	
4	0	30	18	0	0	0	1	1	0	0	12	0	108	
6	0	36	17.5	0	0	0	1	1	0	0	13	0	107	

Figure-10.2-Screenshot of Dataset 2

Batter Runs	Batter Fours	Batter Balls	Batter Total Runs	Dismissal Type	Dismissal Fielder	On Field Minutes	Shot Angle	Shot Magnitude	Wagon Position
1	0	1	1			0	167	104	Backward Square Leg
1	0	1	1			0	252	115	Mid Off
0	0	2	1			0	208	61	Short Extra Cover
4	1	3	5			0	248	201	Long Off
4	2	4	9			0	225	225	Deep Extra Cover
4	3	5	13			0	201	208	Deep Cover
0	0	2	1			0	318	49	Short Extra Cover
6	0	3	7			0	174	197	Deep Square Leg
4	1	4	11			0	124	208	Fine Leg
0	1	4	11			0	90	48	Wicketkeeper
1	1	5	12			0	212	49	Short Mid Wicket
0	3	6	13			0	337	82	Mid Wicket
4	4	7	17			0	145	185	Third Man
6	1	6	18			0	261	211	Long On

Figure-10.3- Screenshot of Dataset 3

Annexure B: Key Performance Indicators (KPI) Definitions

KPI Name	**Formula / Description**
Economy Rate	Runs conceded / Overs bowled
Strike Rate (Batsman)	Balls faced per 100 runs
False Shot %	% of risky or mistimed shots
Dot Ball Rate	% of deliveries where no runs scored
Boundary Frequency	% of balls resulting in 4s or 6s
Match Momentum Index	Score weighted by wickets and over impact

Table-10.4- KPI definition

Annexure C: Visualization Samples

1. Pitch Map – Jos Buttler's Innings against DC
2. Heat zones illustrating scoring with ball lengths (4m–10m)
3. Wagon Wheel – Rinku Singh
4. Visualization of scoring zones in 360-degree view
5. Momentum Graph – KKR vs SRH
6. Animated flow illustrating game turns over 20 overs
7. Feet Movement Analysis

8. Scatter plot illustrating strike rate vs footwork types (front, back, deep)

Annexure D: Tools Used

Tool	Purpose
BallTrack Analyzer	Data extraction and KPI computation
Python (Pandas/Matplotlib)	Data cleaning, analysis, plotting
Flourish Studio	Animated graphs and storytelling visuals
Excel	Initial data inspection and filtering

Table-10.5- Tools

Annexure E: Glossary of Terms

Term	Description
Powerplay	Overs 1–6, with fielding restrictions
Middle Overs	Overs 7–15, focus on consolidation
Death Overs	Overs 16–20, focus on high scoring
Dot Ball	Delivery where no run is scored
Yorker	Ball delivered at the batsman's feet
Back of Length	Ball pitched slightly short of good length

Table-10.6- Terms Glossary